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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU, Belagavi)
IV Semester BE Examinations Oct-2023
Aerospace Engineering
AEROSPACE STRUCTURES

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

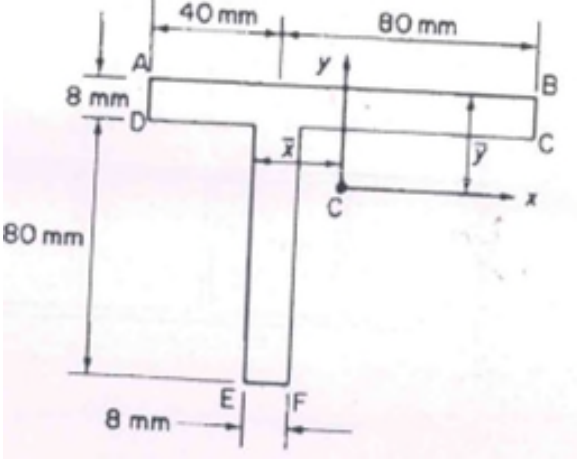
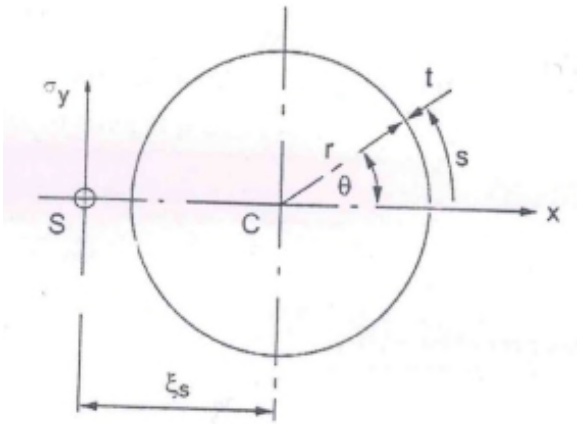
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6, 7 and 8, 9 and 10.

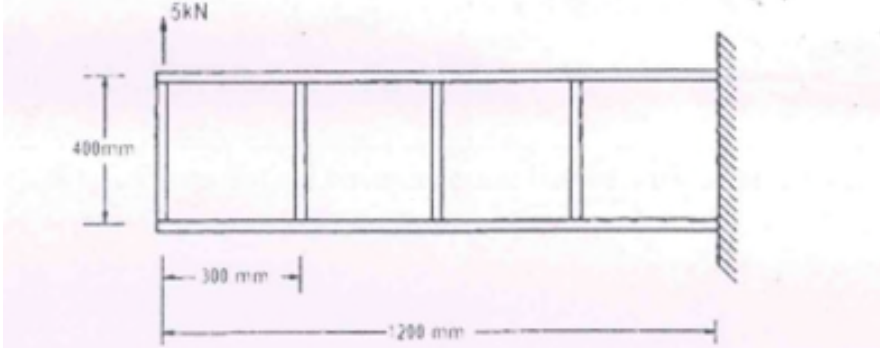
PART A

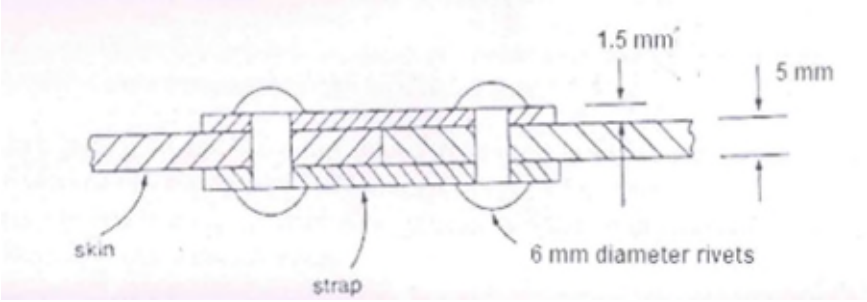
1	1.1	In most of aircrafts where thin shells rely entirely on their skins for their capacity to resist loads, are referred to as _____.	01
	1.2	_____ is a point in the cross-section of open beam through which shear loads produce no twisting.	01
	1.3	_____ factors greater or less than one or even negative are the result of Maneuvers or wind gusts.	01
	1.4	_____ method is used for determining the 'crippling stress' in aircraft structures.	01
	1.5	The elastic _____ stress may be increased by using transverse stiffeners.	01
	1.6	The _____ of an aircraft is predicted on number of flying hours elapsed.	01
	1.7	_____ design philosophy is used by aircraft designers to minimize the effects of metal fatigue and corrosion.	01
	1.8	Torsion box is used in an aircraft wing to provide _____.	01
	1.9	Failures of landing gears are mainly due to _____.	01
	1.10	Resistance welding process is used for welding of _____ in aircraft industries.	01
	1.11	The combined action of _____ and _____ is to resist axial and bending loads.	02
	1.12	What is reduced modulus theory?	02
	1.13	Damage tolerant design philosophy is used by aircraft designers to minimize the effects of _____ and _____.	02
	1.14	List the various failure modes of riveted joints.	02
	1.15	The distance from _____ to the _____ of the weld is called leg.	02

PART B

2	a	Discuss different type of loads acting on aircraft structural components.	06
	b	The wing of a military aircraft has a maximum lift coefficient of 1.25 m and an area of 16m ² . The maximum maneuver load factor is 6.0. If the weight of the aircraft is 50kN, determine the angle bank required at a speed of 180m/s. Also calculate the radius of turn. Take $\rho = 1.223\text{kg/m}^3$	04
	c	Explain briefly the performance of a flight with the aid of flight envelope diagram.	06

3 a	A beam having cross section shown in Fig 3a is subjected to a bending moment of 1500 N-m in a vertical plane. Calculate the maximum direct stress due to bending stating the point at which it acts.  Fig 3a	10 06
b	Discuss shear of open section beams.	06
4 a b	Explain the concept of shear center in closed section beams. Compute the shear flow distribution in the thin walled open section shown in Fig 4b produced by a vertical shear load, σ_y, acting through its shear center. Also sketch the shear flow distribution.  Fig 4b	06 10
5 a b	Derive an equation for critical buckling load of a pin ended column subjected to compressive load. Derive the equation for critical buckling load using the Rankine Gordon Formulation. OR	08 08

6 a b	The beam shown in Fig 6a is assumed to have a complete tension field web. If the cross sectional areas of the flanges and stiffeners are, respectively, 350mm^2 and 300mm^2 and the elastic section modulus of each flange is 750mm^3, determine the maximum stress in a flange and, also check whether the stiffeners will buckle or not? The thickness of the web is 2mm and the second moment of area of a stiffener about an axis in the plane of the web is 2000mm^4. Take $E=70\text{GPa}$.  Fig 6a Calculate the critical load of a column which is made of a bar, circular in cross section and 5m long which is pinned at both ends. The same bar when simply supported gives a mid-span deflection of 10mm with a central concentrated load of 10N.	12 04
7 a b	Explain in detail Damage Tolerant Design by drawing the graph of Residual Strength vs Crack Length. Write a note on life assessment procedures of aircraft structures. OR 8 a b	08 08 08 08
9 a b	With aid of neat diagram, briefly explain the Lug strength analysis under transverse loading condition. What are the advantages and disadvantages of welded joints over riveted joints? OR	08 08

10 a	A joint in a fuselage skin is constructed by riveting the abutting skins between two straps as shown in Fig 10a. The fuselage skins are 2.5mm thick and the straps are each 1.5mm thick: the rivets have a diameter of 6mm. If the tensile stress in the fuselage skin must not exceed 125N/mm^2 and the shear stress in the rivets is limited to 120N/mm^2, determine the maximum allowable rivet spacing such that the joint is equally strong in shear and tension.  Fig 10a	10 06
b	Discuss the welding of aeronautical components and list the various processes of welding.	